Relation:

$A = \{1, 2, 3\}$

$A \times A = \{(1,1), (1,2), (1,3), (2,1), (2,2) \cdots\}$

Axiom of seperation: choice function

I can construct a subset of $A \times A \supseteq R$

say, $A \times A = \{(a,b) : a \in A, b \in A\}$

$R = \{(a,b) : a \in A, b \in A \text{ and } b > a\}$ choice function

$= \{(1,2), (1,3), (2,3)\}$

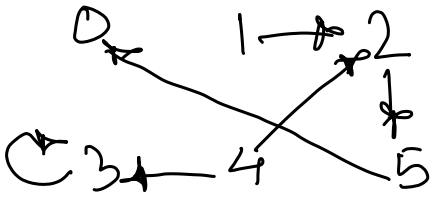
Defn: A relation on a set A is a subset $R \subseteq A \times A$

we often state $(x,y) \in R$ as $x R y$

example: sketch

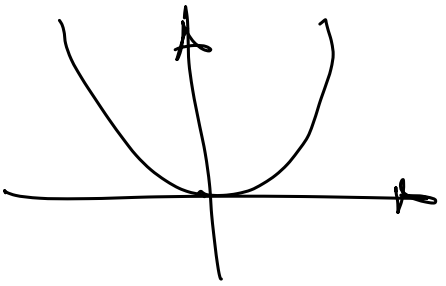
$R = \{(x,y) \in \mathbb{Z} \times \mathbb{Z} : x - y \in \mathbb{N}\}$

(4) a diagram for a Relation R on A



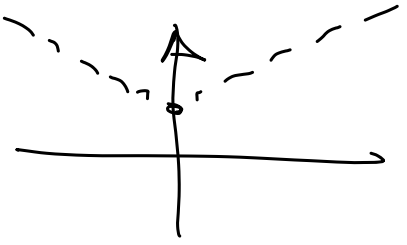
Functions:

$f(x) = x^2 \qquad f: \mathbb{R} \to \mathbb{R}$



$R = \{ (x, x^2) : x \in \mathbb{R} \}$

$f(n) = |n| + 2$



$R = \{ (n, |n| + 2) : n \in \mathbb{Z} \}$

Defn: A & B are sets

A function $f: A \to B$ is a relation, $f \subseteq A \times B$ [from A to B] satisfying: $\forall\, a \in A$, the relation f contains exactly one ordered pair of the form (a, b)

Defn: $f: A \to B$ and $g: A \to B$

Defn: $f: \underbrace{A}_{\text{domain}} \to \underbrace{B}_{\text{codomain}}$

$R = \{ f(a) : a \in A \}$

$R \subseteq$ codomain

$f: \{0,1\} \to \{0,1\}$ $\quad$ $g: \{0,1\} \to \{0,1,2\}$

$f(x) = x$ $\quad$ $g(x) = x^5$

$f(0) = 0$ $\quad$ $g(0) = 0$

$f(1) = 1$ $\quad$ $g(1) = 1$

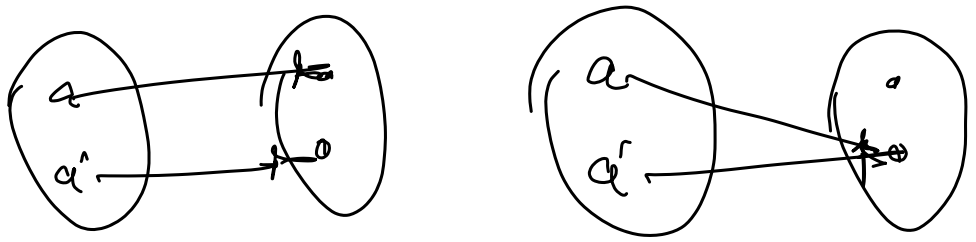
example:

$f: \mathbb{Z} \to \mathbb{N}$ $\quad$ $g: \mathbb{Z} \to \mathbb{Z}$

$f(n) = |n| + 2$ $\quad$ $f(n) = |n| + 2$

Injective function: $f: A \to B$

if $\forall\, a, a' \in A$ $a \neq a'$ implies, $f(a) \neq f(a')$



$f: \{0,1\} \to \{0,1\}$

$f(x) = 0$

How to prove a function is injective

(i) Direct proof

$a, a' \in A$

$a \neq a'$

(ii) Contrapositive proof:

suppose, $a, a' \in A$,

$f(a) = f(a')$

$\Rightarrow f(a) \neq f(a') \quad \Rightarrow \quad a = a'$

$f: \mathbb{N} \to \mathbb{N}$

$f(n) = n^2$ show that this function is one-one!

$n_1, n_2 \in \mathbb{N}$, $n_1 \neq n_2$

$n_1^2 - n_2^2 = \text{sth}$

$\to (n_1 + n_2) \cdot \underbrace{(n_1 - n_2)}_{\neq 0} = \text{sth} \to$ this is not 0

$n_1^2 - n_2^2 = \text{non-zero}$

$n_1^2 = \text{non-zero} + n_2^2$

$\therefore n_1^2 \neq n_2^2$

$f(x) = x^2 + 5 \qquad f: \mathbb{Z} \to \mathbb{Z}$

$x_1, x_2 \in \mathbb{Z} \quad x_1 \neq x_2 \Rightarrow f(x_1) = f(x_2)$

$f(x_1) = x_1^2 + 5 \qquad f(x_2) = x_2^2 + 5$

$f(x_1) - f(x_2) = x_1^2 - x_2^2 = (x_1 + x_2) \cdot (x_1 - x_2)$

since, $x_1 \neq x_2 \quad (x_1 - x_2) \neq 0$

but, $x_1 = -x_2$, $(x_1 + x_2) = 0$

for, $x_1 = -x_2$, since, $x_2 \in \mathbb{Z}$, $-x_2 \in \mathbb{Z}$

$$f(x_1) - f(x_2) = (x_1 + x_2) \cdot (x_1 - x_2) = 0$$

$$\therefore f(x_1) = f(x_2)$$

$$f(x) = x^4 + 2x^3 + 5x + 2 \qquad f: \mathbb{R} \to \mathbb{R}$$

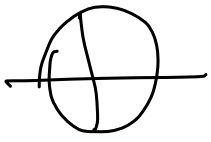
$$x_1, x_2 \in \mathbb{R}, \quad x_1 \neq x_2$$

$$f(x_1) = x_1^4 + 2x_1^3 + 5x_1 + 2$$

$$f(x_2) = x_2^4 + 2x_1^3 + 5x_1 + 2$$

$$f(x_1) - f(x_2) = (x_1^4 - x_2^4) + 2(x_1^3 - x_2^3) + 5(x_1 - x_2)$$

$$= (x_1 - x_2) \boxed{\{(x_1^2 + x_2^2)(x_1 + x_2) + 2(x_1^2 + x_1 x_2 + x_2^3) + 5\}}$$

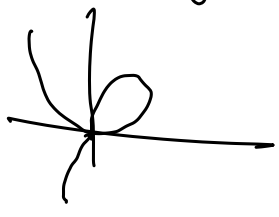


$$x_1 = f(x_2) \qquad x_2 \in \mathbb{R}$$

$$x_1 \in \mathbb{R}?$$

$$x^2 + y^2 = 1 \qquad \ln(xy) + e^x y = 5$$

$$x^3 + y^3 = 3xy$$



Contrapositive:

$x_1, x_2 \in \mathbb{R}$

$$f(x_1) = f(x_2)$$

$$\rightarrow x_1^4 + 2x_1^3 + 5x_1 + \cancel{d} = x_2^4 + 2x_1^3 + 5x_2 + \cancel{d}$$

$$\rightarrow (x_1 - x_2)\{(x_1^2 + x_2^2)(x_1 + x_2) + 2(x_1^2 + x_1x_2 + x_2^2) + 5\} = 0$$

$\therefore x_1 - x_2 = 0$

$\therefore x_1 = x_2$

$x_1^3 + x_1^2x_2 + x_1x_2^2 + x_2^3$
$+ 2x_1^2 + 2x_1x_2 + 2x_2^2$
$+ 5$

$f(x) = e^x \qquad f: \mathbb{R} \rightarrow \mathbb{R}$

is injective?

$x_1, x_2 \in \mathbb{R}, \; x_1 \neq x_2$

$f(x_1) = e^{x_1}$

$f(x_2) = e^{x_2}$

$$e^{x_1} - e^{x_2} = \text{sth}$$

Contrapositive,

$x_1, x_2 \in \mathbb{R},$

$$e^{x_1} = e^{x_2}$$

$$\ln(e^{x_1}) = \ln(e^{x_2})$$

$$\rightarrow x_1 \ln(e) = x_2 \ln(e)$$

$$x_1 = x_2$$

Surjective:

Range = Codomain